

# ISE 5406: Optimization II

## Lecture 2

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### Reading for this lecture

- Chapter 2 of Bazaraa, Sherali, and Shetty (2006)
- Parts of Chapters 2 and 3 of Boyd and Vandenberghe

# Outline

# Definitions: Linear Algebra

- Euclidean Space ( $\mathbb{R}^n$ ): The collection of all vectors of dimension  $n$ .

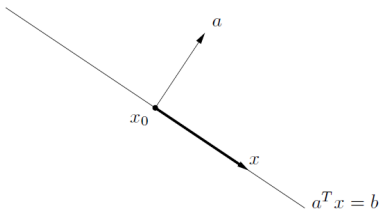
# Definitions: Linear Algebra

- **Euclidean Space ( $\mathbb{R}^n$ ):** The collection of all vectors of dimension  $n$ .
- **Hyperplane:** A set of the form

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} = b\},$$

where  $\mathbf{a} \in \mathbb{R}^n \setminus \{\mathbf{0}\}$  and  $b \in \mathbb{R}$ .

- A line when  $n = 2$  and a plane when  $n = 3$ .



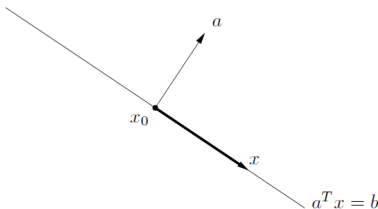
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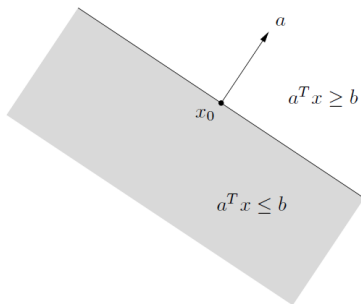
- Set of points  $\mathbf{x}$  with constant inner product ( $b$ ) to given vector  $\mathbf{a}$ .
- A hyperplane divides  $\mathbb{R}^n$  into two **halfspaces**.

# Linear Algebra: Definitions

- Halfspaces:

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \leq b\} \text{ and } \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a}^T \mathbf{x} \geq b\},$$

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# Linear Algebra: Definitions

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# Linear Algebra: Definitions

- **Linear/Affine Subspace** ( $S_L \subseteq \mathbb{R}^n$ ): If  $\mathbf{a}_1$  and  $\mathbf{a}_2 \in S_L$ , then every linear/affine combination of  $\mathbf{a}_1$  and  $\mathbf{a}_2 \in S_L$ .

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- **Span**: Vectors  $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_k$  in  $\mathbb{R}^n$  are said to **span**  $\mathbb{R}^n$  if *any* vector in  $\mathbb{R}^n$  can be represented as their linear combination.

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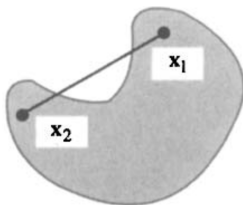
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(a) Convex set



(b) Nonconvex set

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  - 5 Hyperplanes and halfspaces
  - 6  $\{\mathbf{x} \in \mathbb{R}_+^2 : x_1 x_2 = 0\}$
  - 7  $\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 - x_1 x_2^2 + x_2^4 + 1 \geq 0\}$





# Examples of Convex Sets: Polyhedra and Polyhedral Cones

**Polyhedral Set or Polyhedron:** Intersection of a finite number of halfspaces;  $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{b}\}$ , where  $A \in \mathbb{R}^{m \times n}$  and  $\mathbf{b} \in \mathbb{R}^m$ .

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**Polyhedral Cone:** Intersection of a finite number of halfspaces, whose hyperplanes pass through the origin, i.e.,  $\{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \leq \mathbf{0}\}$ .

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**Convex Cone:** A convex set  $X$  with the additional property

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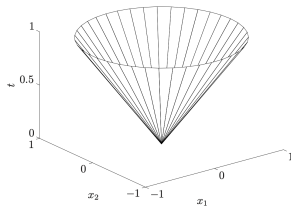
Polyhedral cones are a special case of convex cones.

# Examples of Convex Sets: Ellipsoids and Convex Cones

Examples of convex cones:

- Lorentz or Second-Order Cone:

$$\mathcal{L}^{n+1} := \{(\mathbf{x}, t) \in \mathbb{R}^n \times \mathbb{R} : \|\mathbf{x}\|_2 \leq t\}.$$



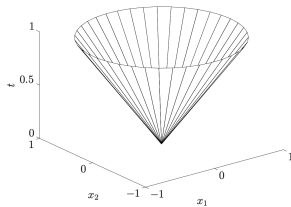
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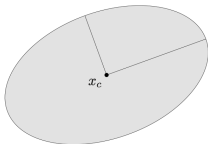


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Other examples of convex sets:

- **Euclidean Ball:**  $\mathcal{B}(\mathbf{x}_c, r) := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{x}_c\|_2 \leq r\}.$
- **Ellipsoid:**  $\mathcal{E} := \{\mathbf{x}_c + A\mathbf{u} : \|\mathbf{u}\|_2 \leq 1\} \subseteq \mathbb{R}^n, \text{ where } A \in \mathcal{S}_+^n.$



# Operations Preserving Convexity of Sets (Not Exhaustive)

## Lemma

Let  $S, S_1, S_2$  be convex sets in  $\mathbb{R}^n$ . Then, the following sets are convex:

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- ⑧ **Affine Image:**  $\{A\mathbf{x} + \mathbf{b} : \mathbf{x} \in S\}$  for any  $A \in \mathbb{R}^{m \times n}$ ,  $\mathbf{b} \in \mathbb{R}^m$ .
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# Convex and Concave Functions

Let  $X \subseteq \mathbb{R}^n$  be a convex set.

**Convex Function:** A function  $f : X \rightarrow \mathbb{R}$  is said to be convex if for *any* two points  $\mathbf{x}_1$  and  $\mathbf{x}_2$  in  $X$  and *any*  $\lambda \in [0, 1]$ ,

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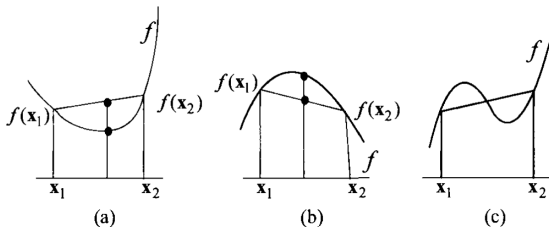
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# Examples of Convex and Concave Functions

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- **Exponential function:**  $\exp(x)$  on  $X = \mathbb{R}$ .
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# Level Sets of Convex Functions are Convex Sets

**(Lower) Level Set:** Let  $S$  be a convex set and  $f : S \rightarrow \mathbb{R}$  be a convex function. Then, for any  $\alpha \in \mathbb{R}$ , the  $\alpha$ -level set of  $f$  is defined as

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## Lemma

The set  $S_\alpha$  is a convex set.

**Proof:**

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- Suppose  $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $i = 1, \dots, m_I$ , are convex and  $h_j : \mathbb{R}^n \rightarrow \mathbb{R}$ ,  $j = 1, \dots, m_E$ , are affine. Then,

$$\{\mathbf{x} \in \mathbb{R}^n : g_i(\mathbf{x}) \leq 0, i = 1, \dots, m_I, h_j(\mathbf{x}) = 0, j = 1, \dots, m_E\}$$

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# Convex Hulls

## Definition

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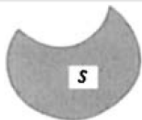
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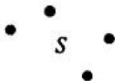
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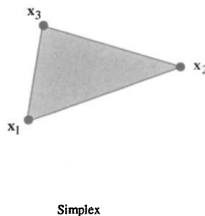
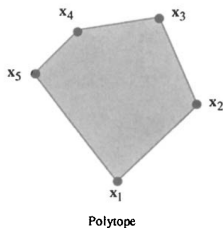
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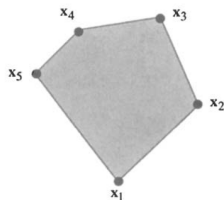
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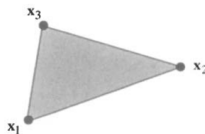
- Let  $S := \{\mathbf{x}_1, \dots, \mathbf{x}_{k+1}\}$  be a finite set of points in  $\mathbb{R}^n$ .
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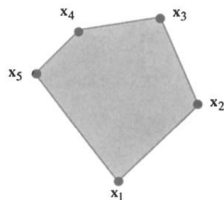
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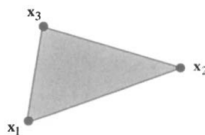
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### Lemma

Let  $S$  be an arbitrary set in  $\mathbb{R}^n$ . Then,  $\text{conv}(S)$  is the smallest convex set containing  $S$ . It is also the intersection of all convex sets containing  $S$ .

# Carathéodory's Theorem (Convex Hull)

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- Trivially holds for  $\mathbf{x} \in S$ .
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# Convex Analysis: Definitions

**Extreme Point:** Let  $X$  be a convex set. A point  $\mathbf{x} \in X$  is called an extreme point of  $X$  if it *cannot be represented* as a *strict convex combination* of two distinct points in  $X$ .

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- Is a ray a convex set?

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**Example:**  $X := \{\mathbf{x} \in \mathbb{R}_+^2 : x_1 - x_2 \geq -2, x_1 + x_2 \geq 1\}$

